DAY 5 — DATASET RESEARCH

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Project Folder: Raritone

Platforms explored: Kaggle, google colab, GitHub

1. Introduction

Datasets play a crucial role in machine learning and artificial intelligence. They are used to train, validate, and test models so that systems can learn patterns and make predictions.

In this task, multiple fashion-related datasets were explored using Kaggle to understand their structure, components, and applications in AI-based systems.

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2. Platforms Used

- Kaggle — Used for exploring and understanding datasets
- Google Colab — Used for basic dataset loading and verification

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3. Datasets Selected

The datasets selected for this task are:

1. Fashion-MNIST Dataset
2. DeepFashion Dataset

DATASET 1 — Fashion-MNIST

4. Dataset Overview

Fashion-MNIST is a dataset of grayscale clothing images used for training machine learning models. It is widely used for image classification tasks and is suitable for beginners.

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5. Dataset Structure

The dataset consists of:

- Training dataset
- Testing dataset
- Labels corresponding to each image

Each image has the following properties:

- Size: 28 × 28 pixels
- Format: Grayscale
- Type: Image data

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6. Sample Dataset Images

Figure 1: Sample images from Fashion-MNIST dataset

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7. Categories

The dataset contains the following categories:

- T-shirt
- Trouser
- Pullover
- Dress

- Coat
- Sandal
- Shirt
- Sneaker
- Bag
- Ankle boot

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8. Observations

- The dataset is simple and well-structured
- Images are easy to process
- Suitable for beginners in AI/ML
- Useful for understanding image classification

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9. Applications

- Clothing classification systems
- Basic machine learning models
- Image recognition systems

DATASET 2 — DeepFashion

10. Dataset Overview

DeepFashion is a large-scale dataset containing real-world clothing images with detailed annotations. It is used for advanced AI applications in fashion technology.

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11. Dataset Structure

The dataset includes:

- High-resolution images
- Attribute labels such as color and style
- Landmark annotations

- Category labels

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12. Sample Dataset Images

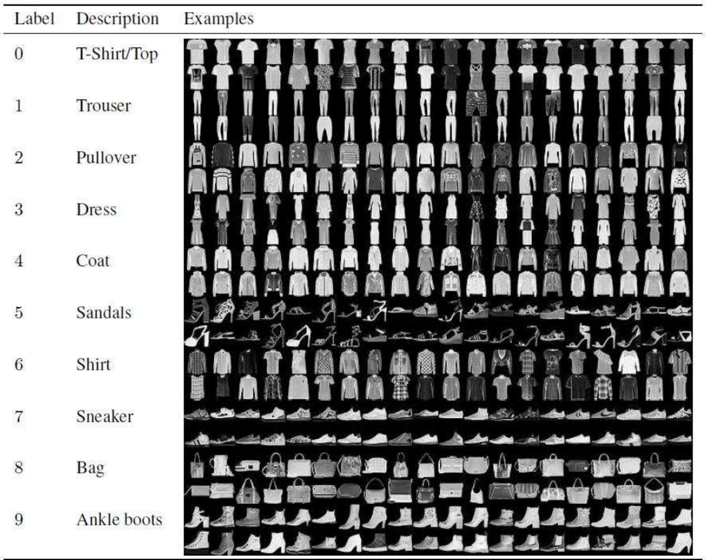

Figure 2: Sample images from DeepFashion dataset

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13. Features

- Real-world clothing images
- Multiple styles and variations
- Complex backgrounds
- Detailed annotations

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14. Observations

- More complex than Fashion-MNIST
- Suitable for advanced models
- Represents real-world scenarios
- Requires preprocessing

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15. Applications

- Virtual try-on systems
- Fashion recommendation systems
- Clothing detection
- Style classification

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16. Comparison of Datasets

Feature	Fashion-MNIST	DeepFashion
Image Type	Grayscale	Color images
Complexity	Low	High
Dataset Size	Small	Large
Use Case	Beginner models	Advanced applications
Real-world Data	No	Yes

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17. Google Colab Demonstration

A basic dataset loading process was demonstrated using Google Colab.

```
from tensorflow.keras.datasets import fashion_mnist

(trainX, trainY), (testX, testY) = fashion_mnist.load_data()

print("Training Data Shape:", trainX.shape)
print("Testing Data Shape:", testX.shape)
```

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18. Observations from Colab

- Dataset loads efficiently
- Contains a large number of images
- Ready for machine learning applications

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19. Conclusion

Both datasets provide valuable insights into how AI systems process image data.

Fashion-MNIST is suitable for beginners and simple classification tasks, while DeepFashion is more complex and suitable for advanced real-world applications.

Exploring multiple datasets helped in understanding the differences between simple and complex datasets and their role in AI-based fashion systems.